Density Operators cant have non-zero elements in all diagonal positions

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LEMMA 1. *A density operator of a pure state cannot have non-zero elements in all diagonal positions.*

PROOF. [1] Density operators are positive operators and so are diagonalizable. Assume the diagonalized density matrix ρ_D. Assume one nonzero entry.

Let $\rho_D = |\psi\rangle\langle\psi|$ for some $|\psi\rangle$. For the state $|\psi\rangle$, let the entries be $x_n, n \in \mathcal{N}$. For any indices i, j such that i $\neq$ j, either x_i is 0 or x_j is zero, since all off diagonal elements are zero. Thus, at least one of x_i and x_j are zero for any distinct pair of indices. $\Rightarrow$ There is only one index i such that $x_i \neq 0 \Rightarrow$ There is only one positive diagonal element in ρ_D

Thus, given the non diagonal elements of a projector are 0 and that the projector is a positive matrix, There can only be one non-zero diagonal element. $\square$

[1]Density operators of a pure state are projectors